CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

November 2013

EC202 Network Theory

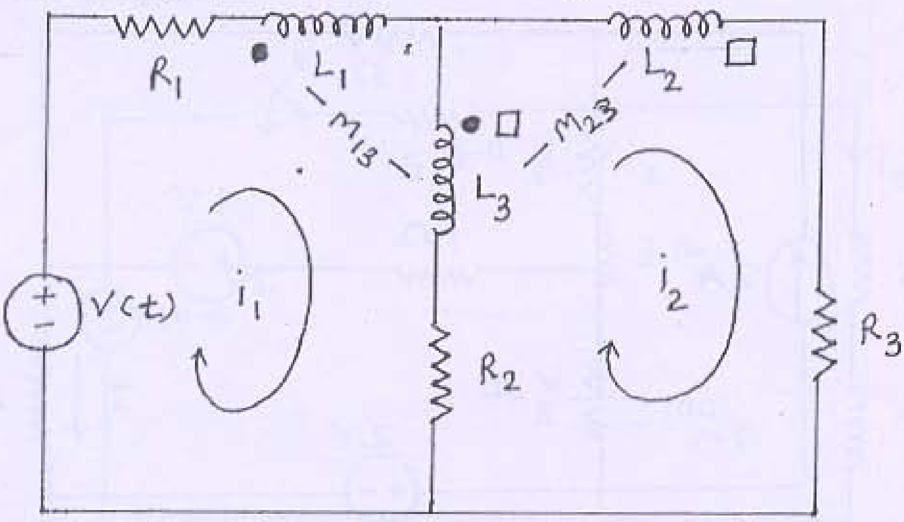
Date: 23.11.2013, Saturday | **Time: 01:30 p.m. To 04:30 p.m.** | **Maximum Marks: 70**

Instructions:

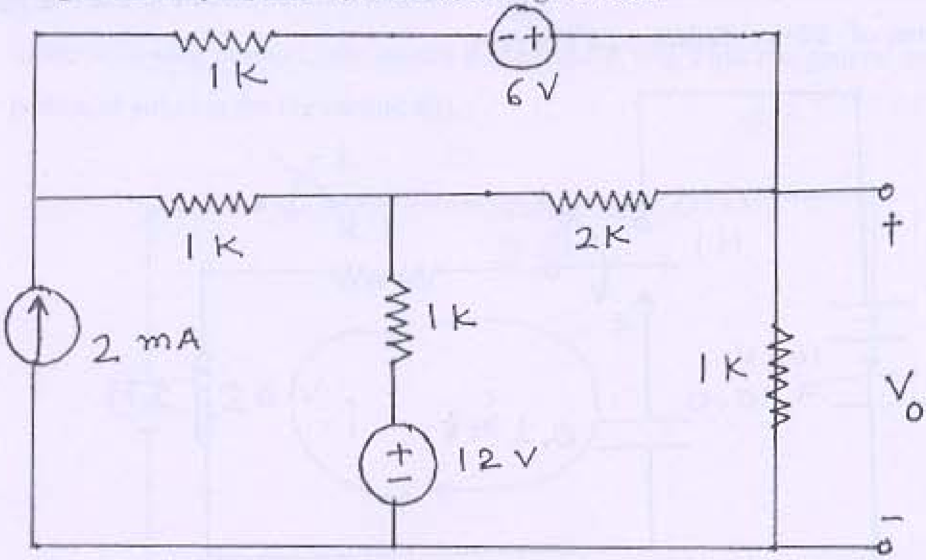
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

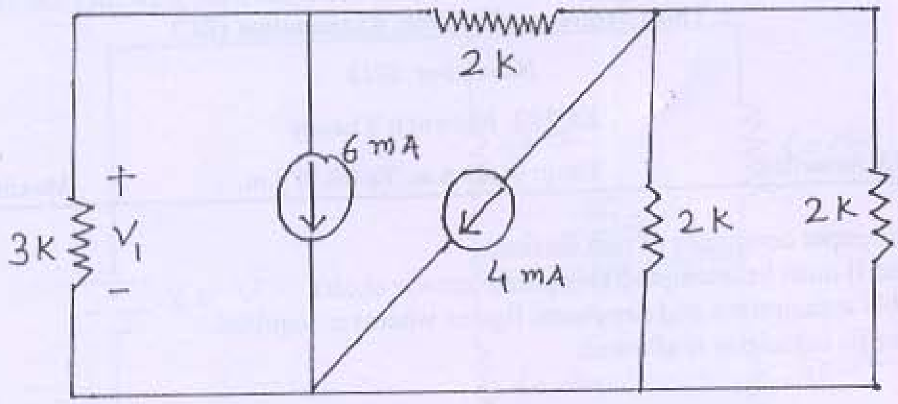
Q - 1 (a) A network with magnetic coupling is shown in following figure, For the network $M_{12}=0$. Write loop equations for this network using KVL. [07]



Q - 2 (a) Using mesh analysis, find V_0 in the following network. [14]

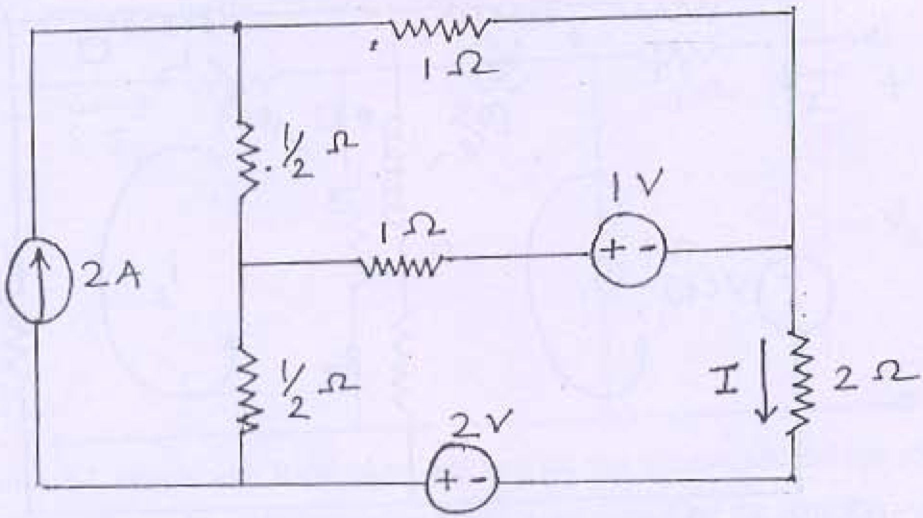


(b) Using node analysis, find V_1 in the following network.

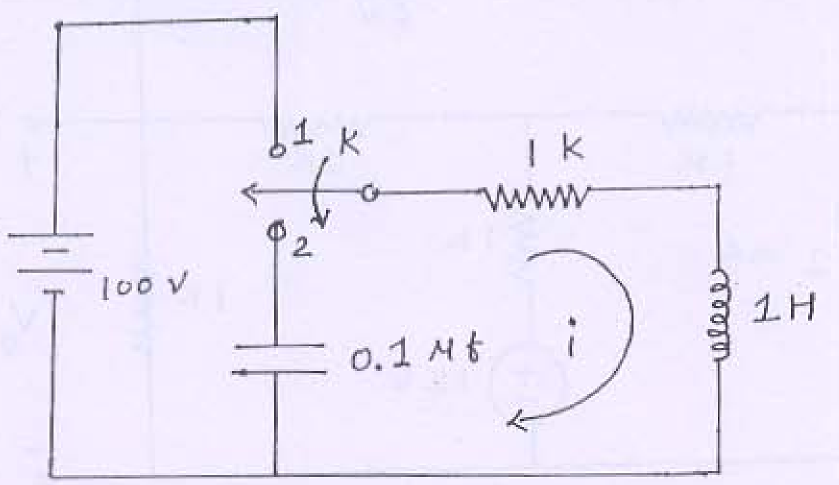


OR

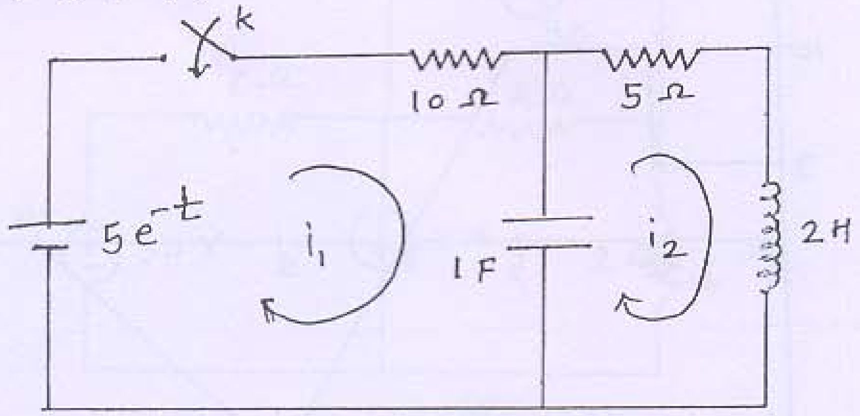
Q - 2 (a) In the following network, Determine the branch current I using source transformation method. [14]



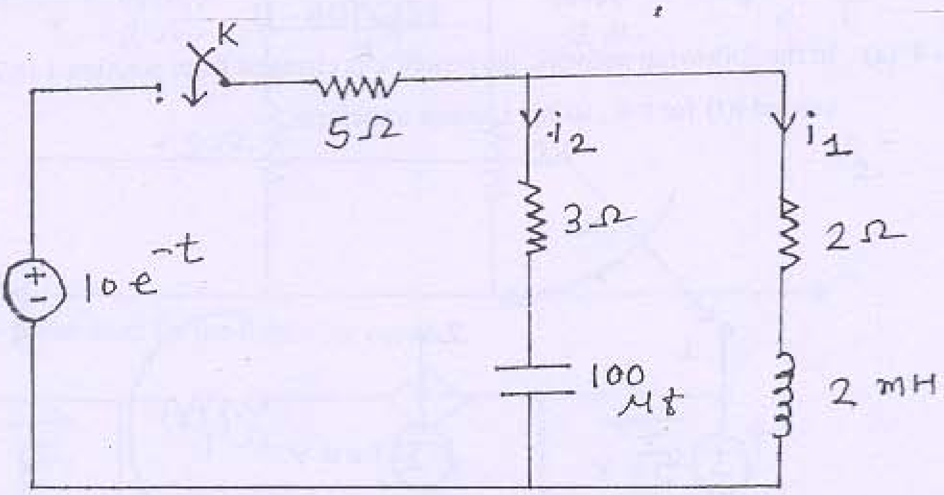
(b) In the following network, the switch k is changed from position 1 to 2 at t=0. Find values of i(0+), di/dt(0+), $d^2i/dt^2(0+)$.



Q - 3 (a) In the following network, the switch k is closed at t=0. Find values of $i_1(0+)$, $i_2(0+)$, $di_1/dt(0+)$, $di_2/dt(0+)$. [14]

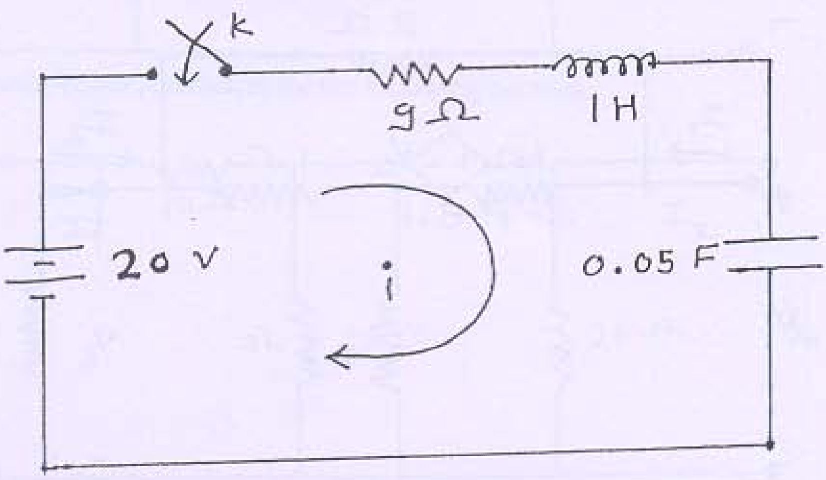


(b) In the following network, the switch k is closed at t=0. Find values of $i_1(0+)$, $i_2(0+)$, $di_1/dt(0+)$, $di_2/dt(0+)$.

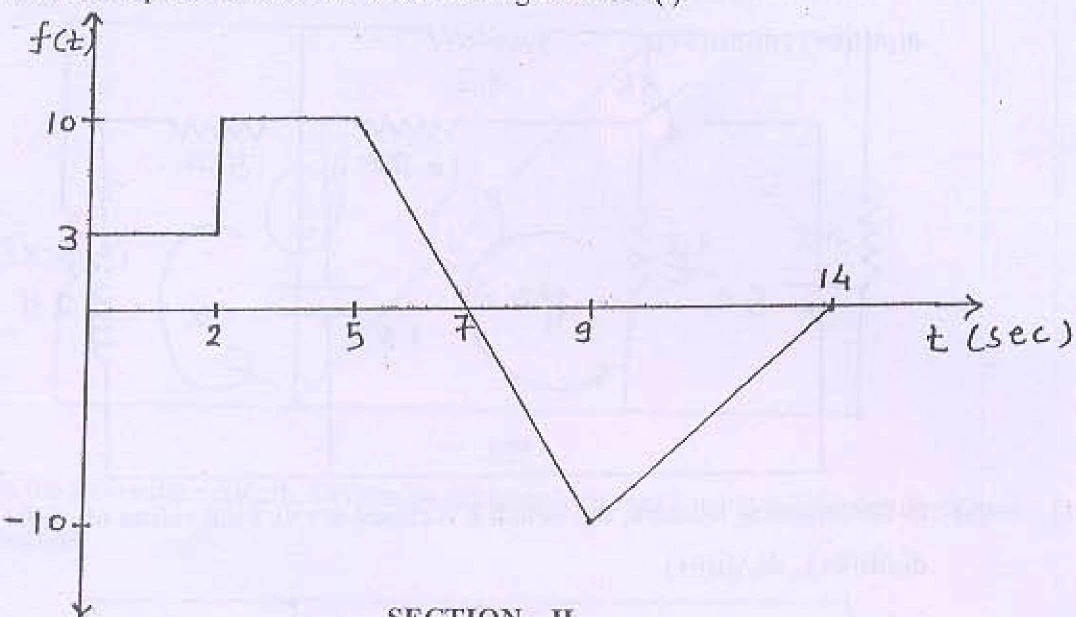


<u>OR</u>

Q - 3 (a) In the following network, the switch k is closed at t=0. Find the general solution and particular solution for the current i(t). [14]

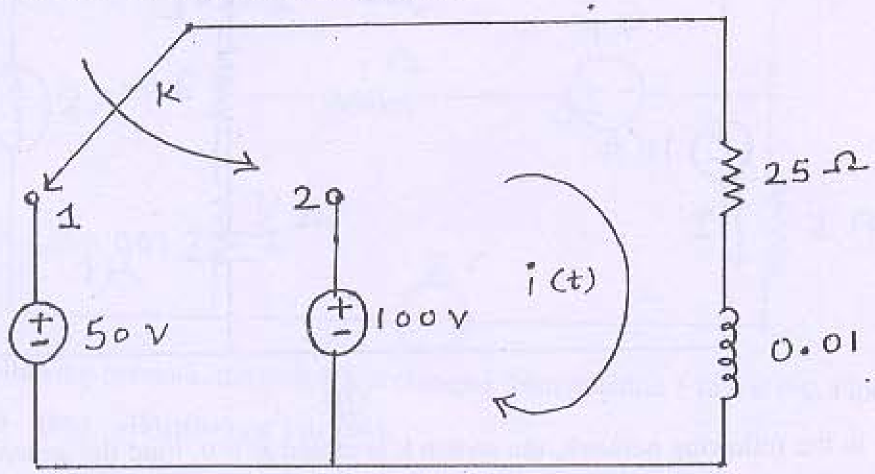


(b) Find the Laplace transform of the following function f(t).

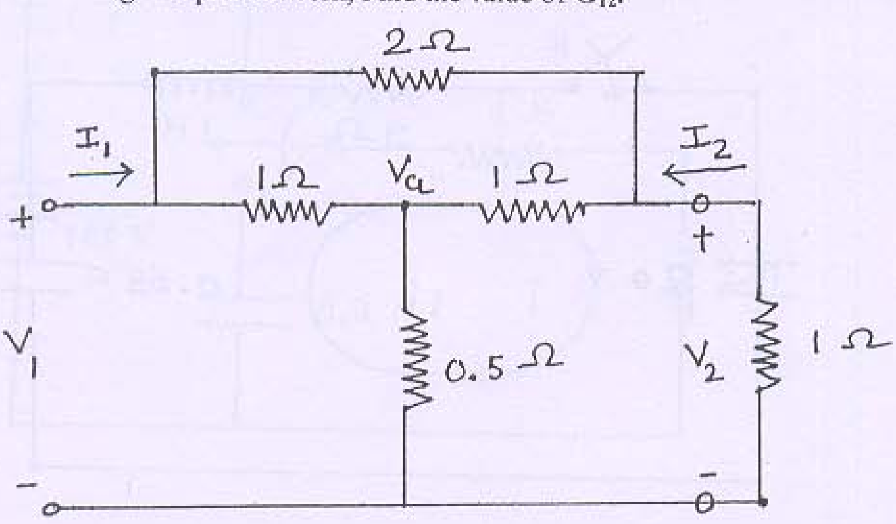


SECTION – II

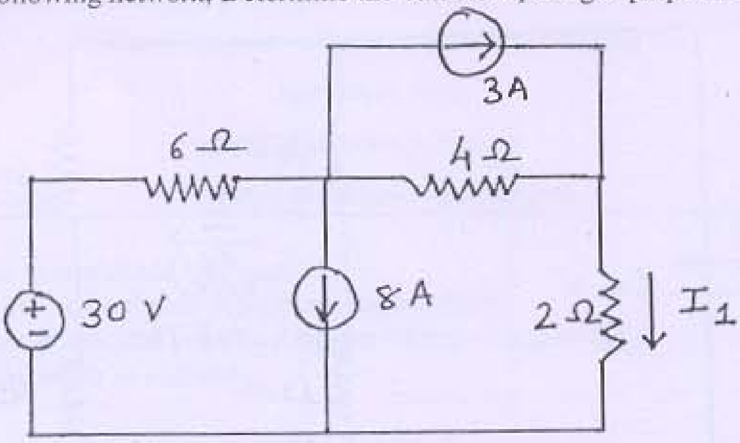
Q - 4 (a) In the following network, the switch k is changed from position 1 to 2 at t=0. Find the current **i(t)** for t>0 , using Laplace transform. **[07]**



Q - 5 (a) In the following two port network, Find the value of $\mathbf{G_{12}}$. **[14]**

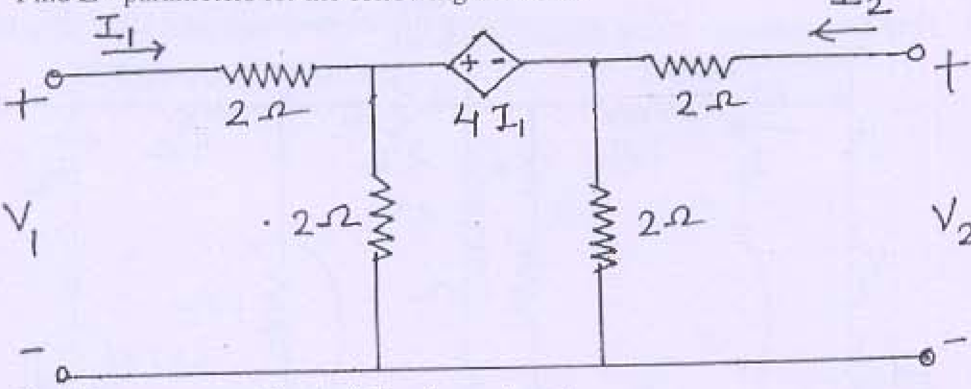


(b) In the following network, Determine the value of I_1 using superposition theorem.

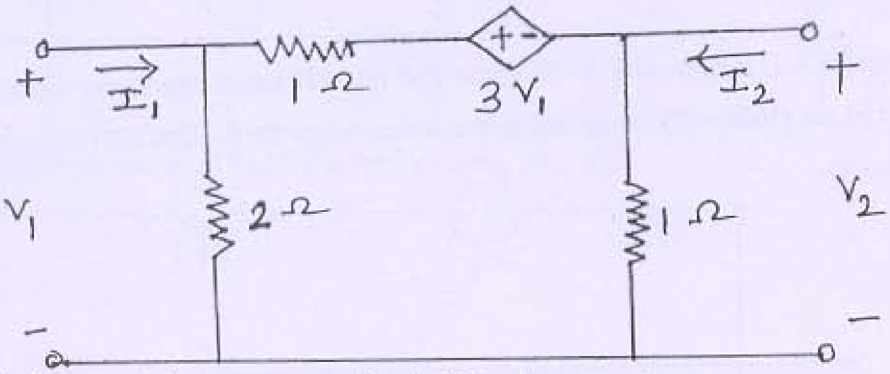


OR

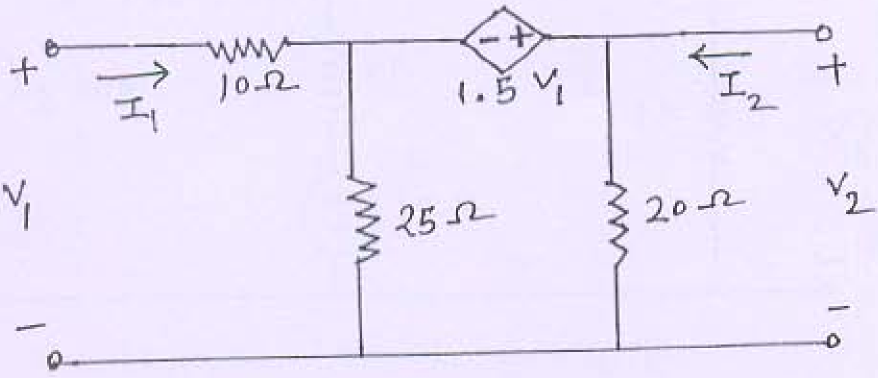
Q - 5 (a) Find Z - parameters for the following network. [14]



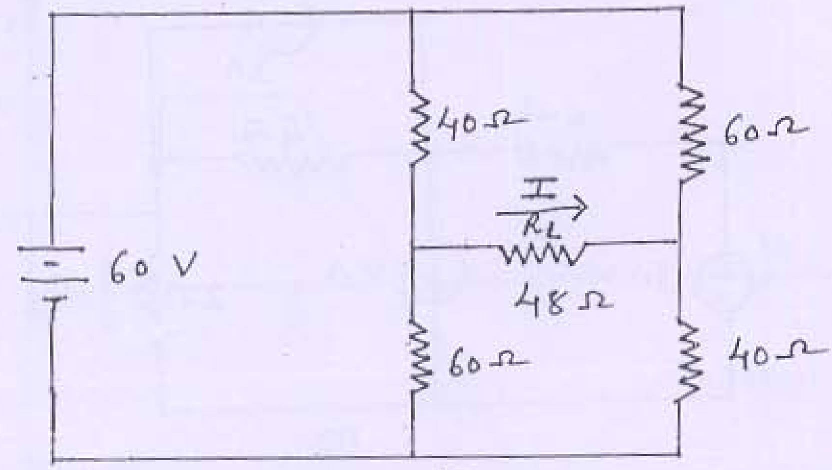
(b) Find Y - parameters for the following network.



Q - 6 (a) Find Transmission parameters for the following network. [14]

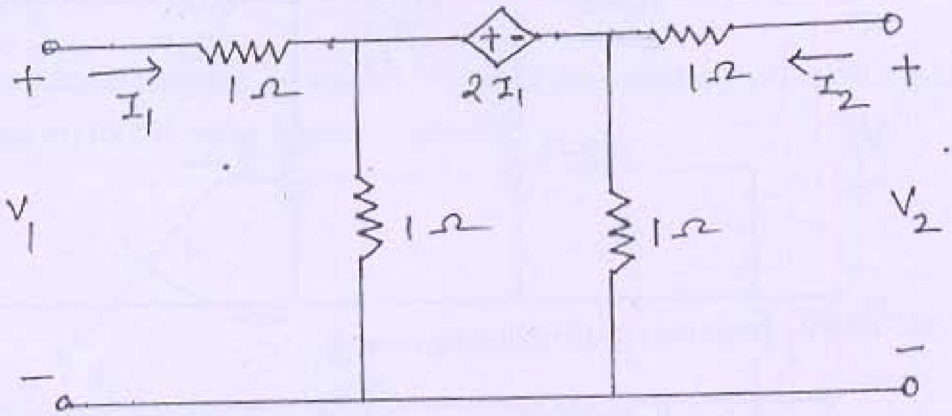


(b) In the following network, Find the current I using Thevenin 's theorem.



OR

Q - 6 (a) Find Y - parameters for the following network. [14]



(b) In series R-L circuit with R=50 ohms and L=0.2H has a sinusoidal voltage source V=150 sin (500t + Θ) volts applied at a time when Θ=0. Find the complete current transient.